

# Quiz 4

---

Matthew Farah

June 30, 2026

## Contents

|                  |           |
|------------------|-----------|
| <b>Problem 1</b> | <b>3</b>  |
| <b>Problem 2</b> | <b>4</b>  |
| (a) .....        | 4         |
| (b) .....        | 4         |
| <b>Problem 3</b> | <b>5</b>  |
| (a) .....        | 5         |
| (b) .....        | 5         |
| (c) .....        | 6         |
| (d) .....        | 7         |
| (e) .....        | 8         |
| <b>Problem 4</b> | <b>9</b>  |
| <b>Problem 5</b> | <b>10</b> |
| <b>Problem 6</b> | <b>11</b> |
| <b>Problem 7</b> | <b>12</b> |
| (a) .....        | 12        |
| (b) .....        | 13        |
| (c) .....        | 13        |
| (d) .....        | 14        |
| (e) .....        | 14        |
| (f) .....        | 14        |
| (g) .....        | 14        |
| <b>Problem 8</b> | <b>15</b> |
| <b>Problem 9</b> | <b>16</b> |

Please note the quiz is based on the following topics:

- Lagrange's Theorem.

**Problem 1:**

Let  $\sim$  be a relation on a set  $S$ . What criteria does  $\sim$  need to satisfy to be an equivalence relation?

**Solution:**

For any relation, including  $\sim$ , to be an equivalence relation, it must satisfy all three of the following criteria:

1. Reflexivity.
2. Symmetry.
3. Transitivity.

**Problem 2:**

Let  $G$  be a group, let  $H \leq G$  and let  $x, y \in G$ .

(a) What is the definition of  $x \equiv y \pmod{H}$  used in this course?

**Solution:**

The definition of  $x \equiv y \pmod{H}$  we've been using so far is that  $x \equiv y \pmod{H}$  if:

$$yx^{-1} \in H$$

(b) What is another, equivalent, definition of  $x \equiv y \pmod{H}$ ?

**Solution:**

As stated before, we define  $x \equiv y \pmod{H}$  to mean:

$$yx^{-1} \in H$$

however, recalling that  $H$  is a subgroup of  $G$ , it must be closed under inversion. Thus,  $yx^{-1} \in H$  if and only if  $(yx^{-1})^{-1} \in H$ . So, since:

$$\begin{array}{ll} yx^{-1} \in H & \text{(By definition of } x \equiv y \pmod{H}\text{)} \\ \iff (yx^{-1})^{-1} \in H & \text{(By closure of } H \text{ under inversion)} \\ \iff (x^{-1})^{-1}y^{-1} \in H & \text{(By exponent laws)} \\ \iff xy^{-1} \in H & \text{(By exponent laws)} \end{array}$$

Thus,  $x \equiv y \pmod{H}$  can be equivalently defined as meaning  $xy^{-1} \in H$ !

As an aside, we also could have arrived at this conclusion by leveraging the fact that congruence modulo  $H$  forms partitions its domain!

Since congruence modulo  $H$  is an equivalence relation, we know it must be symmetric, and by definition of symmetry, we know  $x \equiv y \pmod{H}$  if and only if  $y \equiv x \pmod{H}$ . So, since  $y \equiv x \pmod{H}$ , we know  $xy^{-1} \in H$  (by our usual definition of congruence modulo  $H$ ).

Putting all of this together, we have that  $x \equiv y \pmod{H}$  if and only if  $xy^{-1} \in H$ , which means  $xy^{-1} \in H$  would be an alternative definition for  $x$  being congruent to  $y$  modulo  $H$ .

**Problem 3:**

Let  $G$  be a group, let  $H$  be a subgroup of  $G$ , and let  $\equiv$  denote congruence modulo  $H$ . In each case, determine if the given congruence is correct.

(a) Let  $G = D_4$  and  $H = \langle r \rangle$ . Then  $r \equiv s \pmod{H}$

Note:  $r = (1234)$  and  $s = (13)$ .

**Solution:**

This congruence is incorrect.

To see why, we'll show that  $sr^{-1} \notin H$ . To do that, we'll very quickly compute  $H$ .

We know  $H$  is just the subgroup generated by  $r$ . Since  $r$  is a permutation of order 4, we have that:

$$H = \{ r^0, r^1, r^2, r^3 \}$$

Or alternatively, since  $r = (1234)$ :

$$\begin{aligned} H &= \{ (1234)^0, (1234)^1, (1234)^2, (1234)^3 \} \\ &= \{ (), (1234), (13)(24), (1432) \} \end{aligned}$$

And because:

$$\begin{aligned} sr^{-1} &= (13)(1234)^{-1} \\ &= (13)(4321) \\ &= (14)(23) \end{aligned}$$

$sr^{-1} \notin H$  and so  $r \not\equiv s \pmod{H}$ , by definition of congruence modulo  $H$ .

(b) Let  $G = GL_2(\mathbb{R})$  and  $H = SL_2(\mathbb{R})$ . Then  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \equiv \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \pmod{H}$ .

**Solution:**

This congruence is also incorrect.

For  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  to be congruent to  $\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$  modulo  $H$ , it must be the case that  $\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}^{-1} \in H$ .  
But since:

$$\begin{aligned}
\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} &= \frac{1}{(2)(1) - (1)(0)} \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} && \text{(By definition of inverse)} \\
&= \frac{1}{2} \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} && \text{(Simplifying)} \\
&= \begin{pmatrix} 1 & -1/2 \\ 0 & -1/2 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}$$

We have that:

$$\begin{aligned}
\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}^{-1} &= \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1/2 \\ 0 & -1/2 \end{pmatrix} && \text{(Substituting)} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}$$

Whose determinant is given by:

$$\begin{aligned}
\det \left( \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \right) &= (1)(0) - (1)(1) && \text{(By definition of determinant)} \\
&= 0 - 1 && \text{(Simplifying)} \\
&= -1 && \text{(Simplifying)}
\end{aligned}$$

And thus,  $\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  cannot belong to  $SL_2(\mathbb{R})$ , since  $\det \left( \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \right) = -1 \neq 1$ , and  $SL_2(\mathbb{R})$  only contains matrices with a determinant of 1, by definition.

Hence,  $\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}^{-1} \notin H$ , and so  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  isn't congruent to  $\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$  modulo  $H$ .

(c) Let  $G = GL_2(\mathbb{R})$  and  $H = SL_2(\mathbb{R})$ . Then  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \equiv \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \pmod{H}$ .

**Solution:**

This congruence is actually correct!

As stated before, we know that  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \equiv \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$  if  $\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{-1} \in H$ . And because:

$$\begin{aligned}
\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} &= \frac{1}{(2)(2) - (1)(1)} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} && \text{(By definition of inverse)} \\
&= \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} && \text{(Simplifying)} \\
&= \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}$$

We have that:

$$\begin{aligned}
\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{-1} &= \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix} && \text{(Substituting)} \\
&= \begin{pmatrix} 1/3 & 1/3 \\ -1 & 2 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}$$

And since:

$$\begin{aligned}
\det \left( \begin{pmatrix} 1/3 & 1/3 \\ -1 & 2 \end{pmatrix} \right) &= \left( \frac{1}{3} \right) (2) - \left( \frac{1}{3} \right) (-1) && \text{(By definition of determinant)} \\
&= \frac{2}{3} + \frac{1}{3} && \text{(Simplifying)} \\
&= 1 && \text{(Simplifying)}
\end{aligned}$$

We have that  $\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 1/3 & 1/3 \\ -1 & 2 \end{pmatrix}$  belongs to  $SL_2(\mathbb{R})$  since  $\det \left( \begin{pmatrix} 1/3 & 1/3 \\ -1 & 2 \end{pmatrix} \right) = 1$ , and  $SL_2(\mathbb{R})$  is the set of all  $2 \times 2$  matrices with a determinant of 1 and all real entries, by definition.

Hence,  $\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{-1} \in H$ , and so  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  is congruent to  $\begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$  modulo  $H$ .

(d) Let  $G = S_5$  and  $H = A_5$ . Then  $(123) \equiv (4125) \pmod{H}$ .

**Solution:**

This congruence isn't true.

To see why, recall that  $(123)$  is congruent to  $(4125)$  modulo  $H$  if  $(4125)(123)^{-1} \in H$ . But:

$$\begin{aligned}
 (4125)(123)^{-1} &= (4125)(321) && \text{(Simplifying)} \\
 &= (1354) && \text{(Simplifying)}
 \end{aligned}$$

Which is an odd permutation, since we can express it as the composition of an odd number of length two cycles, like so:

$$(1354) = (13)(35)(1354)$$

And so  $(4125)(123)^{-1} = (1354)$  doesn't belong to  $A_5$ , since  $A_5$  is the set of all even permutations in  $S_5$  definitionally.

Thus, since  $(4125)(123)^{-1} \notin H$ ,  $(123)$  isn't congruent to  $(4125)$  modulo  $H$ .

(e) Let  $G = S_5$  and  $H = A_5$ . Then  $(31)(423) \equiv (2314) \pmod{H}$ .

**Solution:**

This congruence is correct.

Firstly, we can simplify the permutation on the left as follows:

$$(31)(423) = (1342)$$

Meaning we can consider whether or not  $(1342)$  is congruent to  $(2314)$  modulo  $H$  instead! And since:

$$\begin{aligned}
 (2314)(1342)^{-1} &= (2314)(2431) && \text{(Simplifying)} \\
 &= (134) && \text{(Simplifying)}
 \end{aligned}$$

And since  $(134)$  is expressible by:

$$(134) = (13)(34)$$

We know that  $(134)$  is an even permutation, and so  $(2314)((31)(423))^{-1} = (2314)(1342)^{-1} = (134) \in A_5$ , since  $A_5$  is the set of all even permutations in  $S_5$  by definition.

Thus, since  $(2314)((31)(423))^{-1} \in H$ , we have that  $(31)(423)$  is congruent to  $(2314)$  modulo  $H$ .



**Problem 4:**

Let  $G$  be a group and let  $g$  be an element of order 4 and  $x$  be an element of order 5. Let  $H = \langle x \rangle$ . How many elements are in the right coset  $Hg$ ?

**Solution:**

There are exactly 5 elements in the right coset  $Hg$ .

We'll use two different approaches to show this is the case: one by brute force and one appealing to a handful of pretty powerful theorems.

**(I)** The brute force approach.

Starting with the brute force approach, we know that the subgroup generated by  $x$  is representable by the set  $\{1, x, x^2, x^3, x^4\}$ , since  $x$  is an element of order five in  $G$ . Because  $H = \langle x \rangle$ , we therefore have that:

$$H = \{1, x, x^2, x^3, x^4\}$$

Now, letting  $g \in G$ , we know by definition of right cosets that:

$$\begin{aligned} Hg &= \{hg : h \in H\} && \text{(By definition of right cosets)} \\ &= \{(1)g, xg, x^2g, x^3g, x^4g\} && \text{(By iterating over every element in } H) \\ &= \{g, xg, x^2g, x^3g, x^4g\} && \text{(By definition of the identity)} \end{aligned}$$

And thus,  $Hg$  has 5 elements.

**(II)** The argumentative approach.

Recall that, whatever the cardinality of  $Hg$  is, it must equal the cardinality of  $H$  by **Proposition 7.7**. So, finding the number of elements in  $Hg$  can be reduced down to just finding the cardinality of  $H$ !

By **Corollary 4.4**, we know that  $\text{ord}(x) = |\langle x \rangle|$ , meaning that the order of  $x$  equals the order of the subgroup generated by  $x$ . The problem gives us that  $x$  is an element of order five, so we have that  $|\langle x \rangle| = 5$ , and since  $H = \langle x \rangle$ , we have that  $|H| = 5$  as well!

Thus, since  $|H| = |Hg|$  as stated earlier, we can therefore conclude that the right coset  $Hg$  has 5 elements in it.

**Problem 5:**

Which bijection is used in the proof of **Proposition 7.7** from the **Chapter 7** readings?

**Solution:**

The bijection used to prove **Proposition 7.7** was  $f(x) = xa$ .

**Problem 6:**

Fill in the blanks to complete Lagrange's Theorem.

Let  $G$  be \_\_\_\_\_, and let  $H$  be \_\_\_\_\_ of  $G$ . Then  $|H|$  \_\_\_\_\_  $|G|$ .

**Solution:**

Let  $G$  be a finite group, and let  $H$  be a subgroup of  $G$ . Then  $|H|$  is a divisor of  $|G|$ .

**Problem 7:**

Let  $G$  be a group of order 20. Which of the following statements are necessarily true?

(a)  $G$  has an element of order 4.

**Solution:**

This statement is not necessarily true.

The statement likely comes from a misunderstanding of **Corollary 7.9**, which states that if  $x$  is an element of order  $n$  in a group of finite order  $G$ , then  $n$  divides  $|G|$ . Although **Corollary 7.9** is, of course, true, it does not imply that  $G$  must contain an element of the same order of each and every of its divisors.

To give a concrete example that  $G$  doesn't always have an element of order 4, consider the group  $\mathbb{Z}_{10} \times \mathbb{Z}_2$ , which is a group of order 20 which happens not to contain an element of order 4.

Verifying that  $\mathbb{Z}_{10} \times \mathbb{Z}_2$  is indeed a group of order 20 is a relatively trivial endeavour, since:

$$\begin{aligned} |\mathbb{Z}_{10} \times \mathbb{Z}_2| &= |\mathbb{Z}_{10}| |\mathbb{Z}_2| \\ &= (10)(2) \\ &= 20 \end{aligned}$$

Seeing that it doesn't contain an element of order 4, however, is slightly trickier. For  $\mathbb{Z}_{10} \times \mathbb{Z}_2$  to have an element of order 4, then there would need to exist some  $(x, y) \in \mathbb{Z}_{10} \times \mathbb{Z}_2$  such that  $n = 4$  is the smallest positive integer satisfying:

$$n(x, y) = (0, 0)$$

by definition of order, and the fact  $\mathbb{Z}_{10} \times \mathbb{Z}_2$  is an abelian group. By definition of the direct product, the equality above necessarily implies that:

$$(4x, 4y) = (0, 0)$$

By doing a lil' brute force action here, we can see that the only elements in  $\mathbb{Z}_{10}$  satisfying  $4x = 0$  would be  $x = 0$  and  $x = 5$ , since:

$$\begin{array}{ll} (4)(0) \equiv 0 \pmod{10} & (4)(5) \equiv 0 \pmod{10} \\ (4)(1) \equiv 4 \pmod{10} & (4)(6) \equiv 4 \pmod{10} \\ (4)(2) \equiv 8 \pmod{10} & (4)(7) \equiv 8 \pmod{10} \\ (4)(3) \equiv 2 \pmod{10} & (4)(8) \equiv 2 \pmod{10} \\ (4)(4) \equiv 6 \pmod{10} & (4)(9) \equiv 6 \pmod{10} \end{array}$$

In  $\mathbb{Z}_2$  on the other hand, both  $y = 0$  and  $y = 1$  work, since:

$$\begin{array}{ll} (4)(0) \equiv 0 \pmod{2} & (4)(1) \equiv 0 \pmod{2} \end{array}$$

And so, the only possible candidates for an order 4 element in  $\mathbb{Z}_{10} \times \mathbb{Z}_2$  are  $(0, 0)$ ,  $(0, 1)$ ,  $(5, 0)$ , and  $(5, 1)$ . But because:

$$\begin{aligned}
(1)(0,0) &= ((1)(0), (1)(0)) && \text{(By definition of direct product)} \\
&= (0,0) && \text{(Simplifying)} \\
\\
(2)(0,1) &= ((2)(0), (2)(1)) && \text{(By definition of direct product)} \\
&= (0,0) && \text{(Simplifying)} \\
\\
(2)(5,0) &= ((5)(0), (2)(0)) && \text{(By definition of direct product)} \\
&= (0,0) && \text{(Simplifying)} \\
\\
(2)(5,1) &= ((2)(5), (2)(1)) && \text{(By definition of direct product)} \\
&= (0,0) && \text{(Simplifying)}
\end{aligned}$$

that means none of our candidates actually do have an order of 4, since in each case, there exists some integer smaller than 4 which, when multiplied, yields the group's identity.

Therefore,  $\mathbb{Z}_{10} \times \mathbb{Z}_2$  has no element of order 4 and is thus an example of a group of order 20 without such an element.

(b)  $G$  has a subgroup of order 7.

**Solution:**

This statement is always false, and quite easy to disprove at that.

By Lagrange's Theorem, we know that if  $H \leq G$  is a subgroup of  $G$ , then  $|H|$  divides  $|G|$ . However, the order of  $G$  is given to be 20, and 7 isn't a divisor of 20. Thus, by the contrapositive, since any subset of  $G$  with exactly 7 elements would have a cardinality which isn't a divisor of  $|G|$ , no such subset can possibly be a subgroup of  $G$ .

(c)  $G$  does not have a subgroup of index 6.

**Solution:**

This statement is actually true!

Let  $H \leq G$  be an arbitrary subgroup of  $G$ . Recall that the index of  $H$  in  $G$  is defined to be  $|G/H|$ . By **Corollary 7.8**, since  $G$  is a group of finite order, we know  $|G/H| = \frac{|G|}{|H|} = \frac{20}{|H|}$ .

Thus, for the index of  $H$  in  $G$  to be 6,  $H$  would need to satisfy:

$$\begin{aligned}
\frac{20}{|H|} &= 6 \\
\implies |H| &= \frac{20}{6} \\
\implies |H| &= \frac{10}{3}
\end{aligned}$$

Which is clearly impossible, since any group of finite order only has an integer-valued cardinality.

Thus, no subgroup of  $G$  can possibly have an index of 6.

(d)  $G$  has a subgroup of order 4.

**Solution:**

This claim is not necessarily true.

(e)  $G$  does not have an element of order 6.

**Solution:**

The proof of this, once again, follows quickly from **Corollary 7.9**. **Corollary 7.9** tells us that  $|G|$ , which is 20, must be divisible by the order of all of its elements. Since 6 isn't a divisor of  $|G|$ , the contrapositive of **Corollary 7.9** tells us that no element of order 6 can belong to  $G$ .

(f)  $G$  has a nontrivial cyclic subgroup.

**Solution:**

This is actually always the case!

This simply comes from the fact that, since  $G$  is a group of order 20, it must have an element in it that isn't just the identity. Letting  $x \in (G \setminus \{1\})$  be any arbitrary such element, we know that:

1.  $\langle x \rangle$  is a subgroup of  $G$ , since  $x \in G$ .
2.  $\langle x \rangle$  is cyclic, since it is generated by  $x$ .
3.  $\langle x \rangle$  is nontrivial, since  $x \in \langle x \rangle$  and  $x \neq 1$  by definition.

Thus, since we can always find some nontrivial cyclic subgroup of any group of order 20, the claim holds.

(g)  $G$  is cyclic.

**Solution:**

This statement is false.

**Problem 8:**

Find  $[S_7 : D_7]$ .

**Solution:**

$$[S_7 : D_7] = 360.$$

To see how we can arrive at this number, recall that  $|S_n| = n!$  and  $D_k = 2k$  for all integers  $n$  and  $k \in \mathbb{Z}$ . Thus,  $|S_7| = 7! = 5040$  and  $|D_7| = (2)(7) = 14$ . Therefore:

$$\begin{aligned} [S_7 : D_7] &= |S_7/D_7| && \text{(By definition of index)} \\ &= \frac{|S_7|}{|D_7|} && \text{(By \textbf{Corollary 7.8})} \\ &= \frac{5040}{14} && \text{(Substituting)} \\ &= 360 && \text{(Simplifying)} \end{aligned}$$

**Problem 9:**

In the group  $S_4$ , let  $\sigma = (132)$ . What does  $[S_4 : \langle \sigma \rangle]$  equal?

**Solution:**

$$[S_4 : \langle \sigma \rangle] = 8.$$

Here's why. We know that  $\sigma = (132)$  is an element of order 3 in  $S_4$  because:

$$(132)^1 = (132)$$

$$(132)^2 = (123)$$

$$(132)^3 = ()$$

We know then, by **Corollary 4.4**, that  $|\langle \sigma \rangle|$  must equal 3 as well. And since  $S_4$  is evidently a finite group since  $|S_4| = 4! = 24$ , we can call on **Corollary 7.8** to conclude that:

$$\begin{aligned} [S_4 : \langle \sigma \rangle] &= |S_4 / \langle \sigma \rangle| && \text{(By definition of index)} \\ &= \frac{|S_4|}{|\langle \sigma \rangle|} && \text{(By Corollary 7.8)} \\ &= \frac{24}{3} && \text{(Substituting)} \\ &= 8 && \text{(Simplifying)} \end{aligned}$$